

1. GENERAL WELL INFORMATION

Well Name	Alpha-03	Well ID	W003
Field	Field_Alpha	Block	Block_2
Spud Date	14-06-2024	Rig Release Date	18-07-2024
Total Depth (TD)	3350 m	Rig Name	Rig Sagar-4
Well Type	Vertical Development	Trajectory	Vertical
Max Inclination	1.8 deg	Directional Interval	N/A (near-vertical)

2. FORMATION & PORE PRESSURE INFORMATION

Formation_Z (dark grey shale) between 3150 m and 3350 m is over-pressured relative to the regional normal gradient, with pore pressure equivalent mud weight estimated at 1.38-1.44 SG based on post-event analysis. This is the same Formation_Z interval noted in W002 at a shallower structural position, suggesting a field-wide pressure ramp below approximately 3150 m.

3. LITHOLOGY SUMMARY

Interval (m)	Formation	Lithology	Notes
2700-2900	Formation_X (lower)	Sand/shale interbedded	Normal drilling trend
2900-3150	Formation_Y	Shale, grey, firm	Normal pressure trend
3150-3250	Formation_Z (upper)	Shale, dark grey, transitional	Pressure ramp begins; increase gas monitoring
3250-3350	Formation_Z (lower)	Shale, dark grey, over-pressured	Kick risk; primary overpressure zone

4. HISTORICAL DRILLING / OPERATIONS EVENTS

Event 1: Kick (Overpressure Influx) at approximately 3210 m

Observation: Flow was observed on a connection with a pit gain of 6 bbl; background gas increased from 40 units to 210 units over 20 minutes; connection gas peaked at 850 units.

Engineering Action: Well was shut in immediately using the annular BOP (hard shut-in). Shut-in drill pipe pressure (SIDPP) of 380 psi and shut-in casing pressure (SICP) of 460 psi were recorded and logged on the kill sheet. Kill mud weight was calculated at 1.42 SG and the well was circulated out using the Driller's Method across two full circulations.

Outcome: Well was successfully killed with no gas observed at surface after the second circulation. Mud weight was raised from 1.28 SG to 1.42 SG and drilling resumed under close monitoring with connection gas checks on every stand.

5. CASING AND COMPLETION SUMMARY

Casing	Size (in)	Set Depth (m)	Cement Top (m)
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Conductor	20	55	Surface
Surface	13-3/8	640	Surface
Intermediate	9-5/8	2950	1150
Production	7	3350	2500

6. NON-PRODUCTIVE TIME (NPT) SUMMARY

Issue	Depth (m)	NPT (hrs)	Category
Kick / well control event at 3210 m	3210	8.0	Well Control / Overpressure

7. ROOT CAUSE ANALYSIS

Root cause was an underestimated pore pressure gradient in the Formation_Z over-pressured shale; the pre-well pressure prognosis assumed a normal gradient below 3150 m based on regional offset data that did not include this specific pressure ramp. Mud weight at the time of the kick (1.28 SG) was approximately 0.10-0.14 SG underbalanced relative to the actual pore pressure.

8. LESSONS LEARNED

- The Formation_Z pressure ramp begins measurably above 3150 m; connection gas trend (background rising above 100 units) gave roughly 20 minutes of warning before the full kick.
- Kill mud weight of 1.42 SG was sufficient with a trip margin of 0.05 SG; recommend this as the minimum planning MW for offset wells penetrating this interval.
- Two full circulations using the Driller's Method were required; budget well-control contingency time accordingly in future AFEs for this block.
- Real-time pore pressure monitoring (d-exponent / connection gas trending) should be escalated to 'high alert' once background gas exceeds 80 units in this formation.

9. RECOMMENDATION FOR OFFSET WELLS

For any offset well drilling into Formation_Z below 3150 m in Block_2, plan mud weight at a minimum of 1.40-1.44 SG before entering the interval, increase connection gas monitoring frequency, and pre-stage kill-weight mud (1.45 SG) on surface ahead of penetrating 3150 m.